JOHN ZHANG

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EDUCATION

UNIVERSITY OF TORONTO Toronto, ON
Bachelor of Science Expected Jun 2028
Major in Computer Science; Major in Statistical Science
Cumulative GPA: 3.89/4.0; Innis College Anniversary Scholarship 2025, Dean's List Scholar 2025
Relevant Coursework: Machine Learning, Data Analysis, Statistics, Deep Learning, Data Structures & Algorithms

WORK EXPERIENCE

SUNNYBROOK RESEARCH INSTITUTE Toronto, ON
Machine Learning Analyst May 2025 – Present

- Engineered deep learning models (CNN, LSTM) to translate data between modern and legacy accelerometers, enabling reuse of historical actigraphy data in clinical research and contributing to an upcoming publication.
- Developed and evaluated machine learning models for Alzheimer's disease prediction using EEG and sleep-stage data, including feature engineering and subject-level validation.
- Designed and containerized a scalable sleep–wake classification pipeline with Docker for reproducible analysis of large-scale UK Biobank actigraphy data.

UNIVERSITY OF TORONTO AEROSPACE TEAM Toronto, ON
Data Processing Engineer Sep 2024 – Jan 2025

- Debugged and modernized legacy Python and SQL code while researching hyperspectral image correction methods, restoring satellite-data processing capabilities and improving image accuracy for scientific analysis.

PROJECTS

AUTOML AGENT Live Demo | GitHub Repo | Jul 2026 – Present

- Architected a multi-agent LLM pipeline (LangGraph) that turns a natural-language ML request into a trained model, automating dataset analysis, experiment planning, model generation and result reporting end-to-end.
- Built a retrieval-augmented research agent that retrieves relevant excerpts from a Qdrant vector store using the request and dataset profile, grounding the planning agent throughout the experimentation loop.
- Engineered a self-repairing experimentation agent that generates library-agnostic Python training code, installs dependencies in an isolated environment, and automatically diagnoses and fixes runtime failures.
- Deployed on GCP (Cloud Run, Firestore/Cloud Storage, Firebase) with live prediction serving, drift monitoring (Evidently), one-click retraining, and a LangSmith eval suite in CI to catch agent regressions.

COST-AWARE TOOL-USE LEARNING FOR LLM GitHub Repo | Sep 2026 – Present

- Built a cost-aware tool-use agent environment in Python with 35 priced tools in 10 overlapping cost families (e.g., single vs. batch APIs) and Pydantic-typed state. An LLM selects multi-step tool calls as structured JSON.
- Built a synthetic data pipeline that executes every valid and invalid path of 7.7K tasks from 50 templates, producing 11K SFT demonstrations and 2.8K step-level DPO preference pairs with leakage-free splits.
- Fine-tuned Qwen2.5-1.5B with LoRA SFT, then DPO (TRL) on an H100 via Slurm. SFT raised held-out task success from 0% to 98%; DPO cut spending above the cheapest valid path from 29% to 11%.
- Built a two-stage evaluation harness (GPU inference, then LLM-as-judge) with cost-efficiency metrics. Error analysis traced DPO's 5-pt success drop to imbalanced preference pairs and unrecovered tool failures.

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face (Transformers, PEFT, TRL), LoRA/QLoRA Fine-Tuning, DPO, LLM Evaluation (LLM-as-Judge, LangSmith), Evidently, TensorBoard
Engineering: Python, Pydantic, FastAPI, LangGraph, RAG, Qdrant, Docker, Git, CI/CD (GitHub Actions)
Data/Cloud: NumPy, Pandas, SciPy, SQL, Google Cloud Platform, Signal Processing